NAICHENG JIANG

Stockholm, Sweden | naicheng@kth.se | +46 76 748 35 21
jackson-nicholas.github.io

RESEARCH INTERESTS

3D generative modelling and spatial representations; vision-language understanding; reliable multimodal AI, including applications in health.

EDUCATION

3+2 bachelor-master study pathway.

KTH Royal Institute of Technology, Stockholm, Sweden
MSc in Information and Network Engineering | Aug 2025 - Jun 2027 (expected)
Weighted grade average: 4.72/5.00 (self-calculated from completed graded courses).

Shanghai Jiao Tong University, Shanghai, China
BSc in Electrical and Computer Engineering | Aug 2022 - Aug 2026
UM-SJTU Joint Institute (Global College) | GPA: 3.57/4.00

HONORS AND AWARDS

Outstanding Graduate Award, Undergraduate Class of 2026 - Shanghai Jiao Tong University
Undergraduate Academic Scholarship, Class C, Shanghai Jiao Tong University (2023 and 2024)
Yu Liming Scholarship, UM-SJTU Joint Institute (2024)

RESEARCH EXPERIENCE

VecSetX - Compositional 3D Generation from Incomplete Supervision
Collaborative research project | Feb 2026 - Present
Conditional completion and unconditional generation of 3D anatomy from four partially annotated datasets.

- Extended and fine-tuned a multi-source 3D tokenizer for latent generative modelling, achieving 0.923 mean reconstruction Dice across 204 validation cases.
- Prototyped a conditional diffusion transformer with cross-attention; designed context-shuffling controls to test evidence use and diagnose conditioning failures.
- Evaluated cross-source generalization and geometric consistency through baseline comparisons, distinguishing structural plausibility from patient-specific accuracy.

CURRENT AND PLANNED PROJECTS

Flow Matching for Image Reconstruction and Translation | KTH EQ2443
Ongoing course project | Oct 2026 - Present

- Designing a U-Net velocity-field baseline for flow-matching reconstruction from perturbed images, with image-to-image translation as a further research objective.
- Planning controlled evaluations of reconstruction fidelity, flow-estimation accuracy and sampling cost, using image-quality metrics and runtime to study quality-compute trade-offs.

3D Vision-Language Understanding
Proposed research
Research interest in grounding language-based reasoning in volumetric visual evidence, with emphasis on spatial relationships and consistency between generated responses and observed anatomy.

TECHNICAL SKILLS

Python, MATLAB, C++; PyTorch, OpenCV, NumPy/SciPy; Linux, Git, LaTeX, Slurm, LUMI/HPC, CUDA.

LANGUAGES

Chinese (native); English (professional working proficiency).